Literature status: endpoint questions from arXiv:1712.01374

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Status. `literature_already_answered`. This note records later literature and makes no claim of a new proof.

Source questions. Randrianantoanina–Wu–Xu, *Noncommutative Davis type decompositions and applications*, arXiv:1712.01374, asks in Remark 3.11 whether its symmetric-space Davis decomposition extends to every $E \in \mathrm{Int}(L_1, L_q)$, $1 < q < \infty$, and whether

$$\mathcal{H}_E^c(\mathcal{M}) = \mathsf{h}_E^d(\mathcal{M}) + \mathsf{h}_E^c(\mathcal{M})$$

holds for every $E \in \mathrm{Int}(L_1, L_2)$. The remark following its Burkholder/Rosenthal theorem also asks whether either usual sum/intersection formula applies to $L_{2,\infty}$, or more generally $L_{2,q}$ with $q \neq 2$.

Explicit Davis answer. Randrianantoanina, *P. Jones' interpolation theorem for noncommutative martingale Hardy spaces*, arXiv:2212.08714, explicitly cites the source as `Ran-Wu-Xu` and states that it resolves the problem left there. Proposition `Davis-symmetric`(i) proves, for every $E \in \mathrm{Int}(L_1, L_q)$,

$$E^{\mathrm{ad}}(\mathcal{M}; \ell_2^c) \subseteq E\left(\bigoplus_{n \geq 1} \mathcal{M}_n\right) + E^{\mathrm{cond,ad}}(\mathcal{M}; \ell_2^c),$$

which is the requested endpoint decomposition. Part (ii) and Corollary `Davis` give the displayed Hardy-space identity for every $E \in \mathrm{Int}(L_1, L_2)$. The same source proves this threshold sharp: validity of the column Davis identity forces $E \in \mathrm{Int}(L_1, L_2)$.

Weak-L_2 answer. Cadilhac, *Noncommutative Khintchine inequalities in interpolation spaces of L_p-spaces*, arXiv:1809.06091, Proposition `counterexample` and Section 6, constructs finite Schur–Horn examples showing that neither standard row/column Khintchine formula holds in $L_{2,\infty}$. Either Burkholder/Rosenthal alternative asked for in the source would imply the corresponding Rademacher/Khintchine estimate; hence the specifically highlighted weak-L_2 case has a negative answer. Cadilhac's arXiv:1812.03861, Theorem `khrad`, gives the subsequent strict-interpolation characterization of both standard Rademacher formulas.

Provenance. The 2022 paper knows that it is answering Remark 3.11 of the source. The weak-L_2 conclusion is a direct consequence of the later explicit Khintchine counterexamples and of the necessary reduction already noted in the source paper.

References

[1] N. Randrianantoanina, L. Wu, and Q. Xu, *Noncommutative Davis type decompositions and applications*, arXiv:1712.01374.

[2] L. Cadilhac, *Noncommutative Khintchine inequalities in interpolation spaces of L_p-spaces*, arXiv:1809.06091.

[3] L. Cadilhac, *Majorization, interpolation and noncommutative Khintchine inequalities*, arXiv:1812.03861.

[4] N. Randrianantoanina, *P. Jones' interpolation theorem for noncommutative martingale Hardy spaces*, arXiv:2212.08714.